Install Maven:

Open CMD Prompt and check the Java Version:

```
Windows PowerShell
Windows PowerShell
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Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Windows\System32\WindowsPowerShell\v1.0> java --version
openjdk 21.0.5 2024-10-15 LTS
OpenJDK Runtime Environment Temurin-21.0.5+11 (build 21.0.5+11-LTS)
OpenJDK 64-Bit Server VM Temurin-21.0.5+11 (build 21.0.5+11-LTS, mixed mode, sharing)
PS C:\Windows\System32\WindowsPowerShell\v1.0>
```
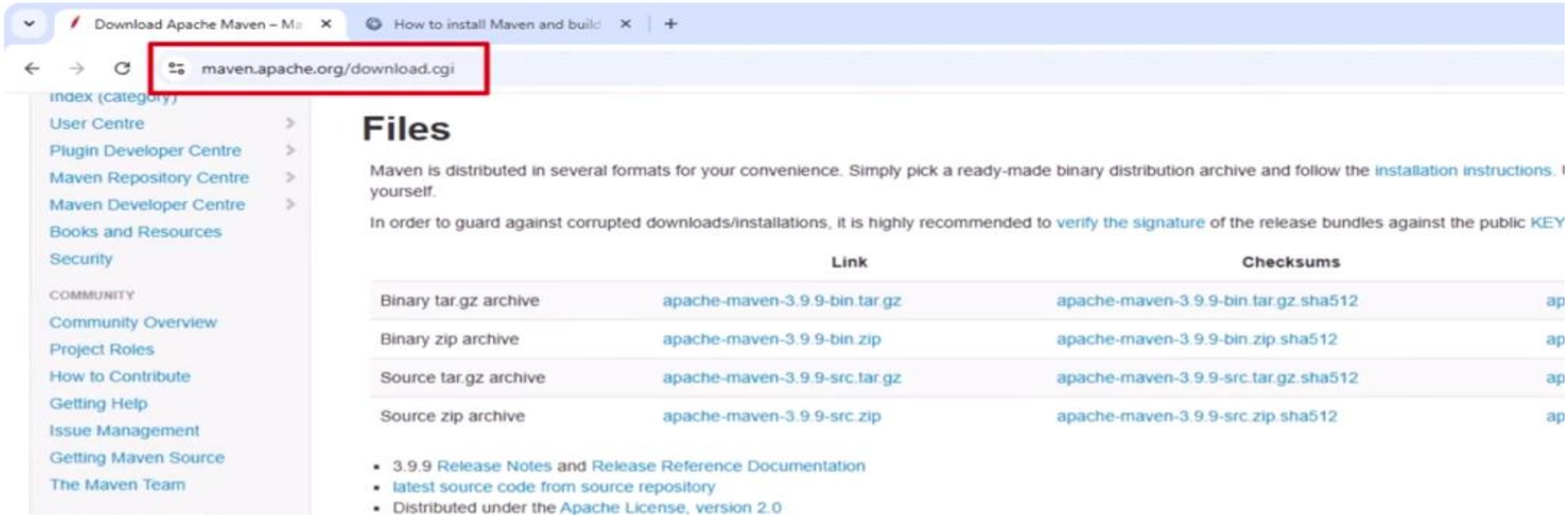
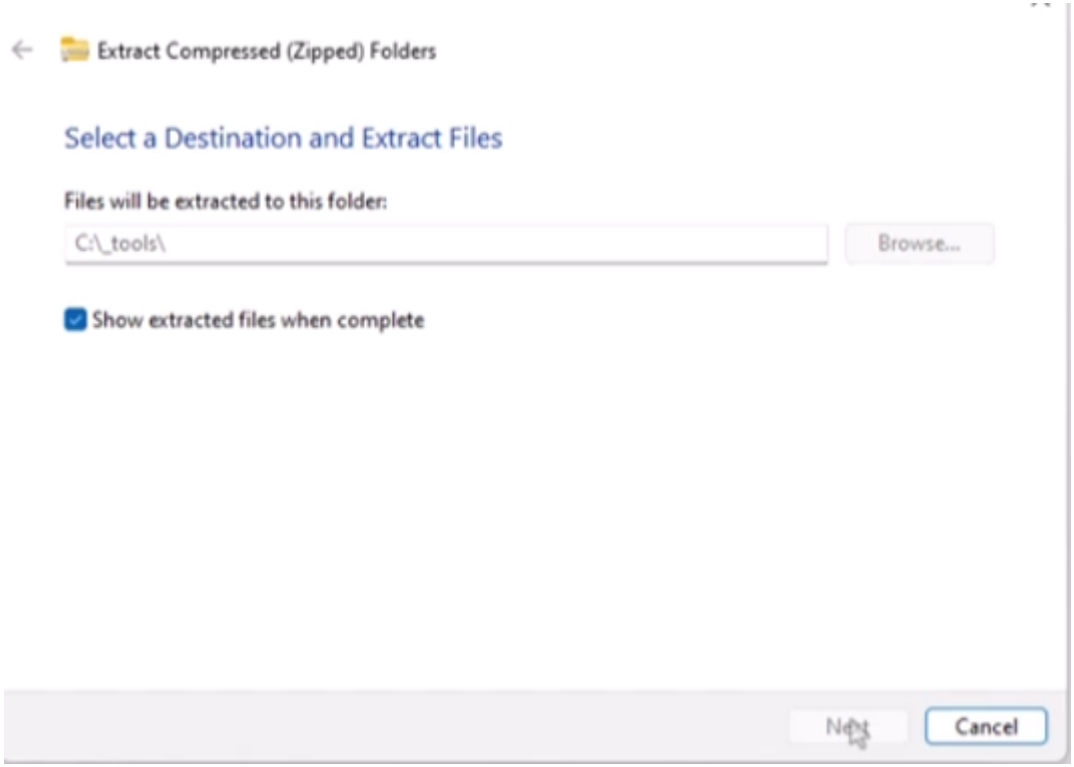
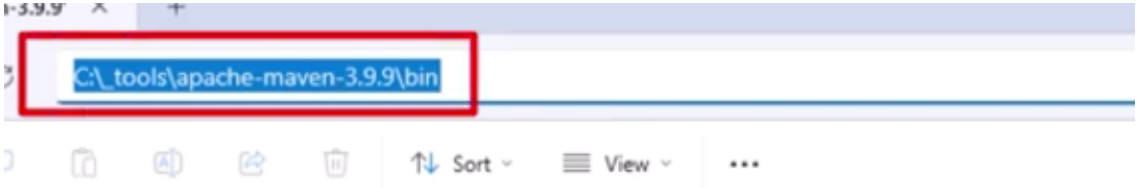

Go to APACHE MAVEN: https://maven.apache.org/download.cgi



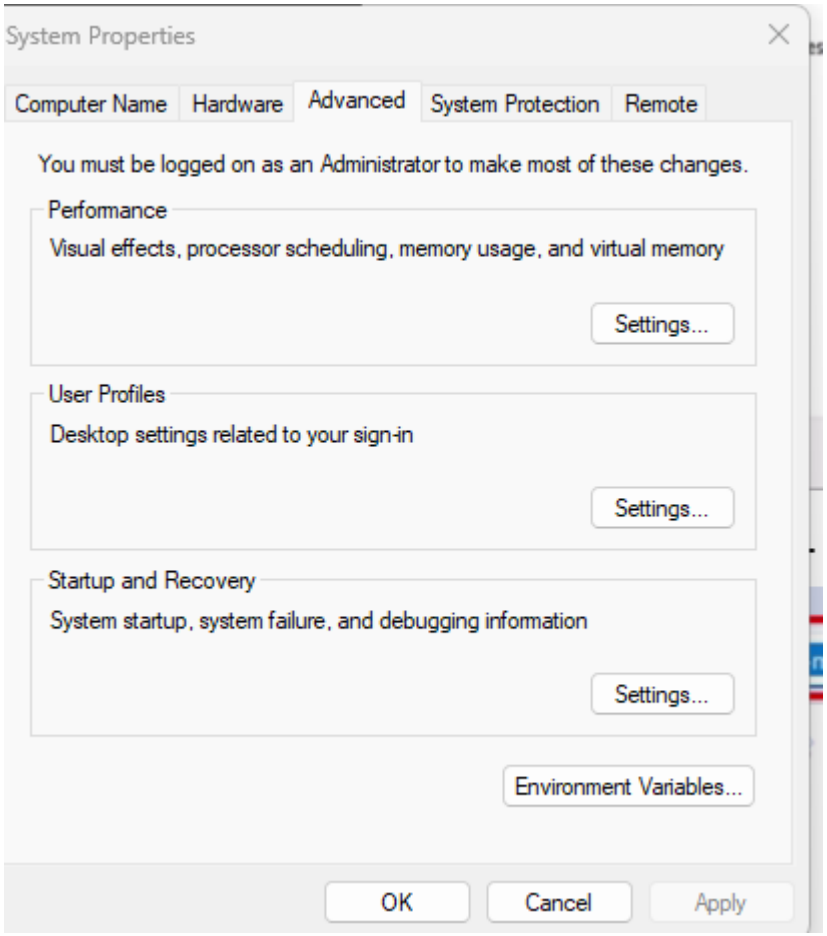
1. **DOWNLOAD APACHE-MAVEN BIN.ZIP**
2. **SAVE IT UNDER THE SAME FOLDER THAT YOU HAVE YOUR JDK INSTALLED**
3. **EXTRACT ALL IN THAT SAME FOLDER**



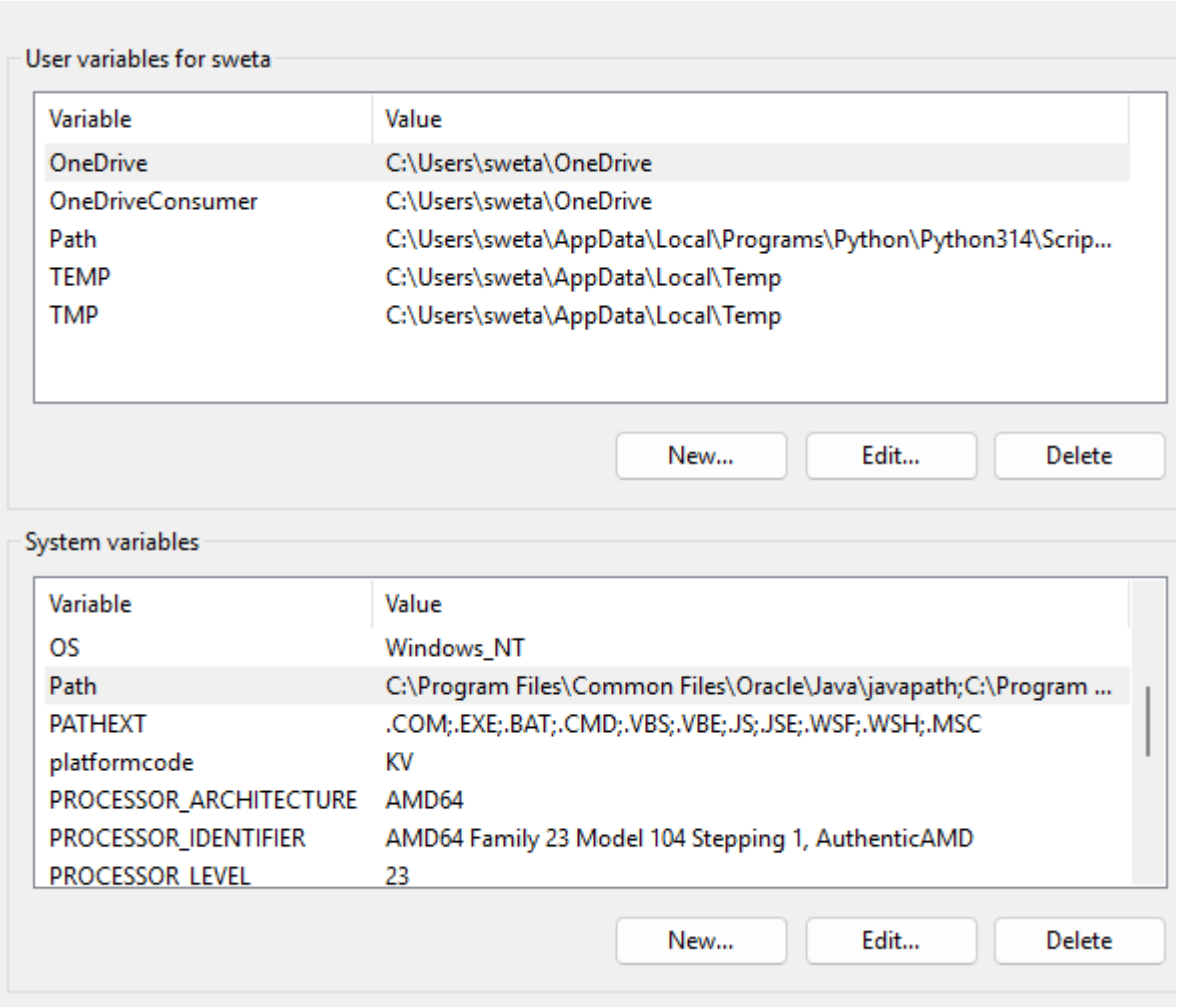
4. **COPY FULL PATH OF THE BIN DIRECTORY**



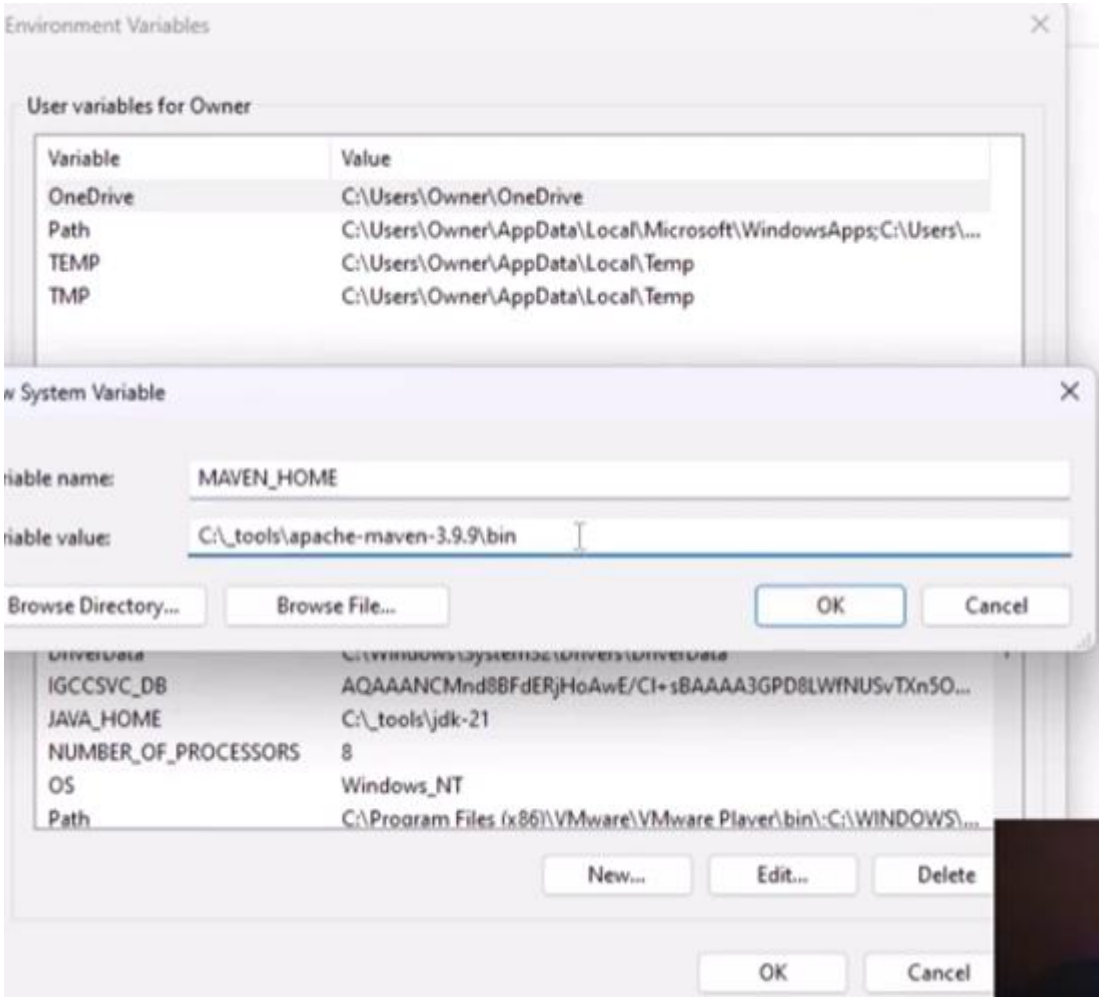
5. **EDIT THE ENVIRONMENT VARIABLE**

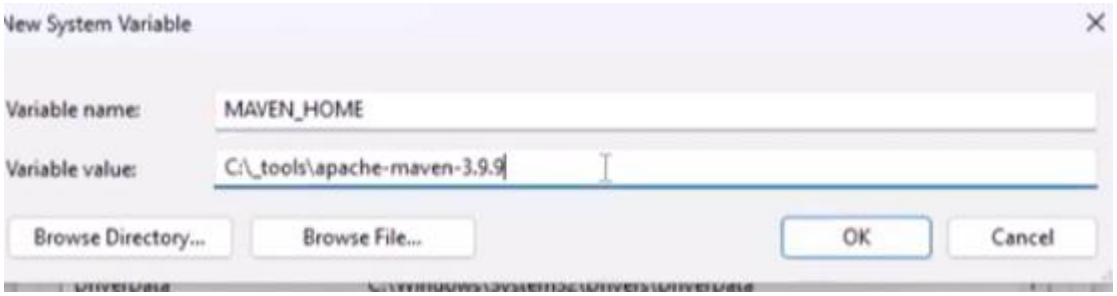


6. **CLICK ON ENVIRONMENT VARIABLE**
7. **CLICK ON PATH AND EDIT:**

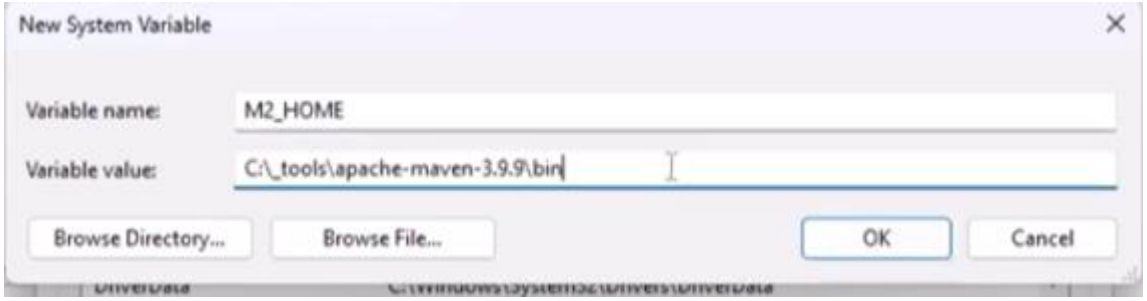


8. **CLICK ON NEW AND PASTE THE PATH OF THE MAVEN BIN AND CLICK ON OK**
9. **Then ON THE ENVIRONMENT VARIABLE SCREEN CLICK ON NEW → ENTER VARIABLE NAME AS MAVEN_HOME, VARIABLE VALUE: PASTE THE SAME LINK AND REMOVE BIN FROM THAT LINK**

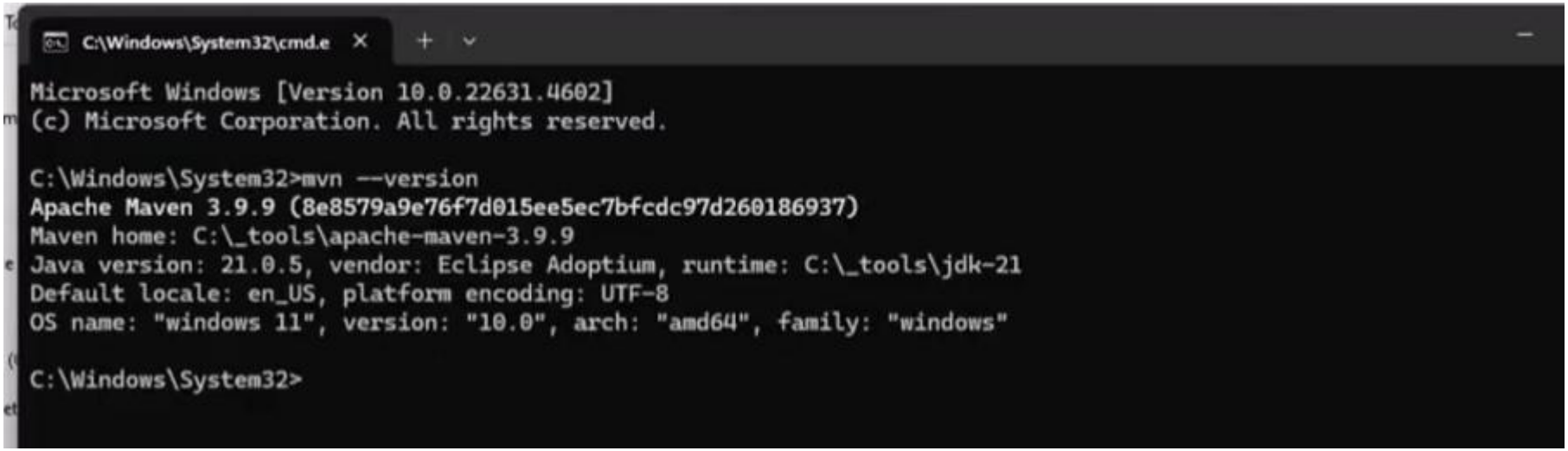

Environment Variables
User variables for Owner
Variable
Value
OneDrive
C:\Users\Owner\OneDrive
Path
C:\Users\Owner\AppData\Local\Microsoft\WindowsApps;C:\Users\...
TEMP
C:\Users\Owner\AppData\Local\Temp
TMP
C:\Users\Owner\AppData\Local\Temp
System Variable
MAVEN_HOME
C:_tools\apache-maven-3.9.9\bin
Browse Directory...
Browse File...
OK
Cancel
IGCCSVC_DB
AQAAANCMnd8BFdERjHoAwE/CI+sBAAAA3GPD8LWfNUSvTXn5O...
JAVA_HOME
C:_tools\jdk-21
NUMBER_OF_PROCESSORS
8
OS
Windows_NT
Path
C:\Program Files (x86)\VMware\VMware Player\bin\;C:\WINDOWS\...
New...
Edit...
Delete
OK
Cancel


New System Variable
Variable name:
MAVEN_HOME
Variable value:
C:_tools\apache-maven-3.9.9
Browse Directory...
Browse File...
OK
Cancel

SAME WAY AGAIN CLICK ON NEW AND ADD VARIABLE NAME AND VALUE AS FOLLOWS – REMOVE BIN FROM THE VARIAVLE VALUE LINK:



GO BACK TO COMMAND PROMPT AND TYPE mvn --version

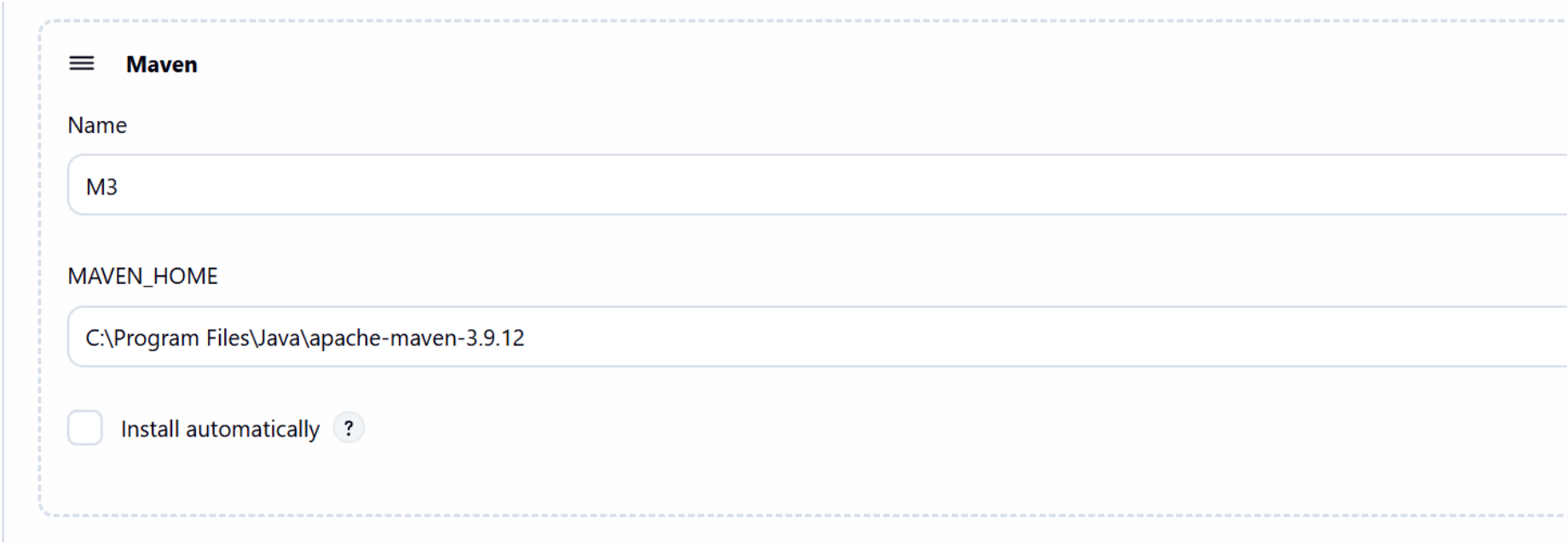


SO MAVEN IS SUCCESSFULLY INSTALED.

CONFIGURE MAVEN IN JENKINS:

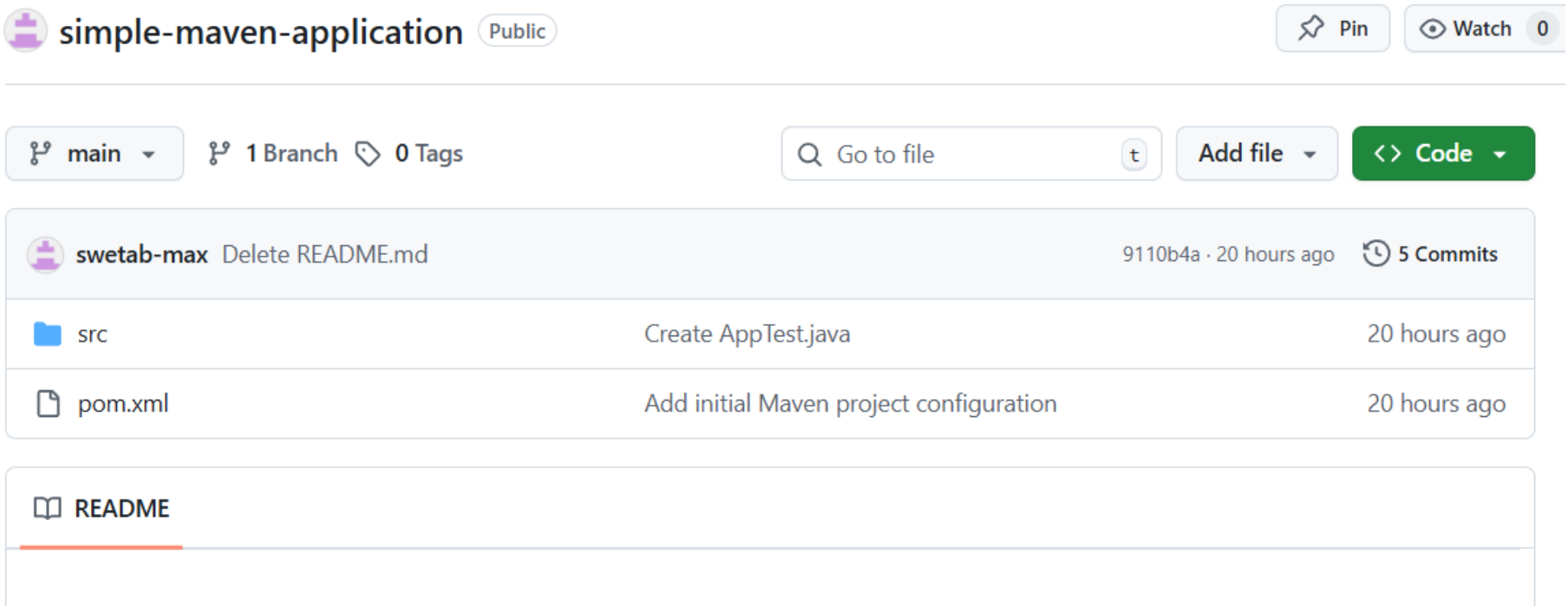
Path:
Manage Jenkins → Tools → Maven Installations

Configuration:
Name: M3
MAVEN_HOME: C:\Program Files\Apache\apache-maven-3.9.12
Install Automatically: UNCHECKED



NOW LETS OPEN GITHUB

1. **CREATE A NEW REPOSITORY: SIMPLE-MAVEN-APPLICATION**



Create a file – pom.xml

main

SIMPLE-MAVENAPP2 / pom.xml

swetab-max Add initial Maven project configuration in pom.xml

Code Blame 18 lines (17 loc) · 561 Bytes

```xml
<project xmlns="http://maven.apache.org/POM/4.0.0"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.example</groupId>
 <artifactId>simple-maven-app</artifactId>
 <version>1.0-SNAPSHOT</version>

 <dependencies>
   <dependency>
       <groupId>junit</groupId>
       <artifactId>junit</artifactId>
       <version>4.13.2</version>
       <scope>test</scope>
   </dependency>
 </dependencies>
</project>
```

CODE:

```
<project xmlns="http://maven.apache.org/POM/4.0.0"
 xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
 xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
 http://maven.apache.org/xsd/maven-4.0.0.xsd">
 <modelVersion>4.0.0</modelVersion>
 <groupId>com.example</groupId>
 <artifactId>simple-maven-app</artifactId>
 <version>1.0-SNAPSHOT</version>

 <dependencies>
  <dependency>
     <groupId>junit</groupId>
     <artifactId>junit</artifactId>
     <version>4.13.2</version>
     <scope>test</scope>
  </dependency>
 </dependencies>
</project>
```

COMMIT THE CHANGES

THEN CREATE THE FOLLOWING FOLDER STRUCTURE

```
└── src/
    ├── main/java/com/example/App.java
    └── test/java/com/example/AppTest.java
```

\

FOR THE APP.JAVA PLEASE USE THE FOLLOWING CODE:

```
package com.example;
public class App {
 public int add(int a, int b) {
 return a + b;
 }
}
```

COMMIT THE CHANGES

FOR THE AppTest.JAVA use the FOLLOWING CODE:

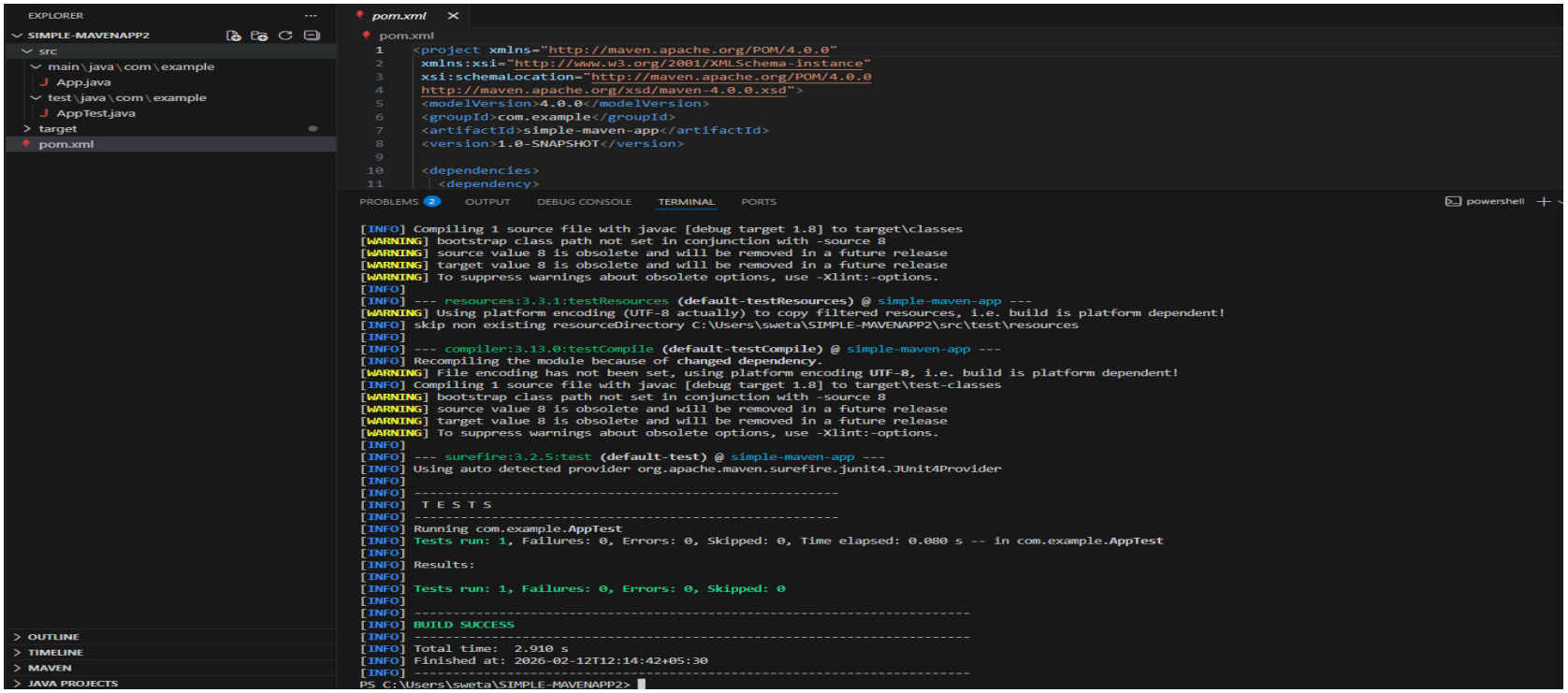
```
package com.example;
import org.junit.Test;
import static org.junit.Assert.*;
public class AppTest {
 @Test
 public void testAdd() {
 App app = new App();
 assertEquals(5, app.add(2, 3));
```

```
 }
}
```

COMMIT THE CHANGES

THEN OPEN VSCODE:

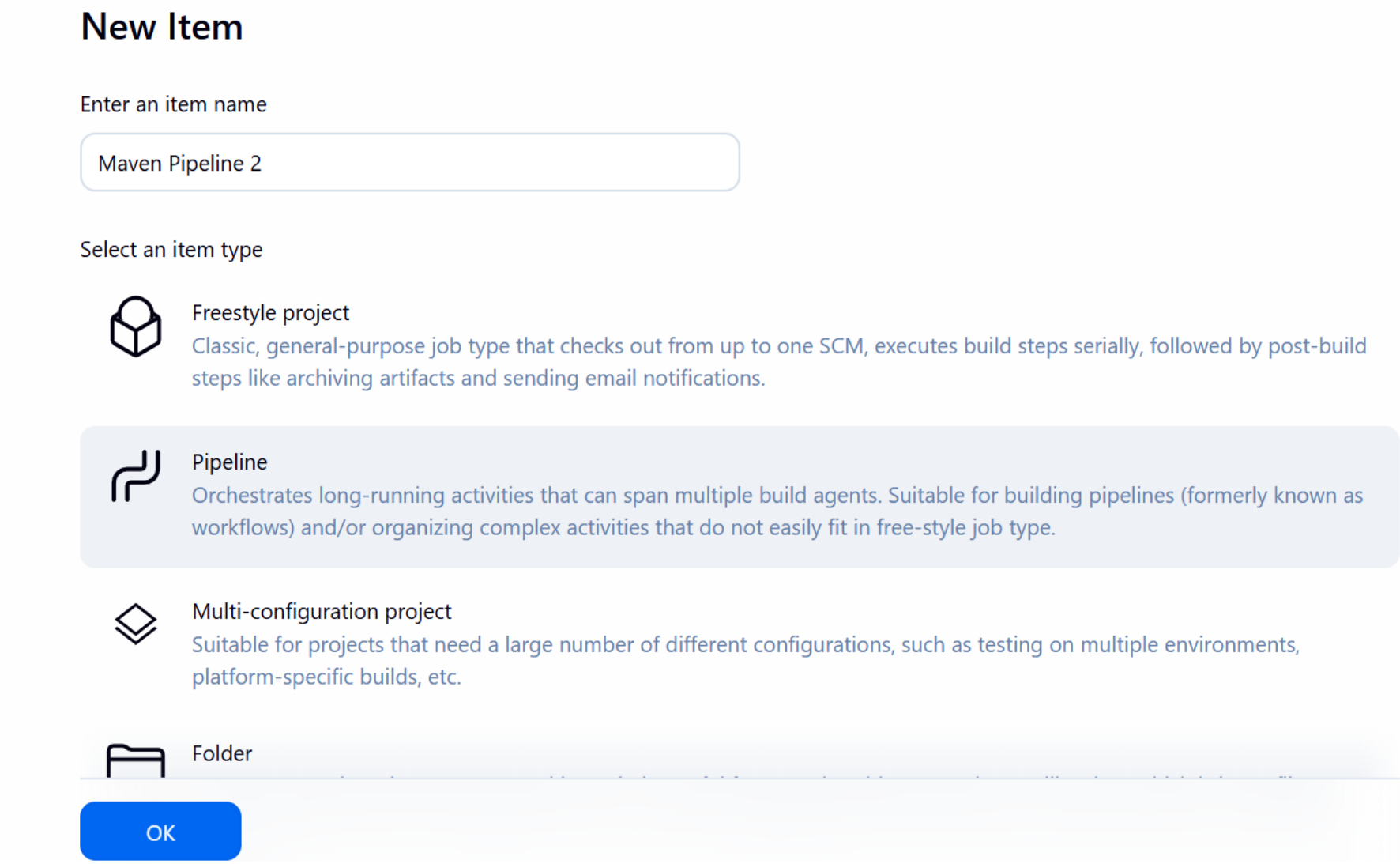
1. **CLONE YOUR GIT REPOSITORY IN VSCODE**
2. **In the TERMINAL type command: mvn clean test**



IT SHOULD SHOW THAT BUILD IS SUCCESSFUL

NOW OPEN JENKINS:

1. **NEW ITEM**
2. **ENTER NAME AND SELECT PIPELINE**
3. **CLICK OK**



IN THE PIPELINE SCRIPT ADD THE FOLLOWING CODE:

```
pipeline {
 agent any
 tools {
 maven 'M3'
 }
 stages {
 stage('Checkout Git') {
 steps {
 git branch: 'main',
 url: 'https://github.com/balaganeshn/simple-maven-app'
 }
 }
 stage('Build and Test') {
 steps {
 bat 'mvn clean test'
 }
 }
 }
}
```

UPDATE THE GITHUB url TO YOUR REPOSITORY URL:

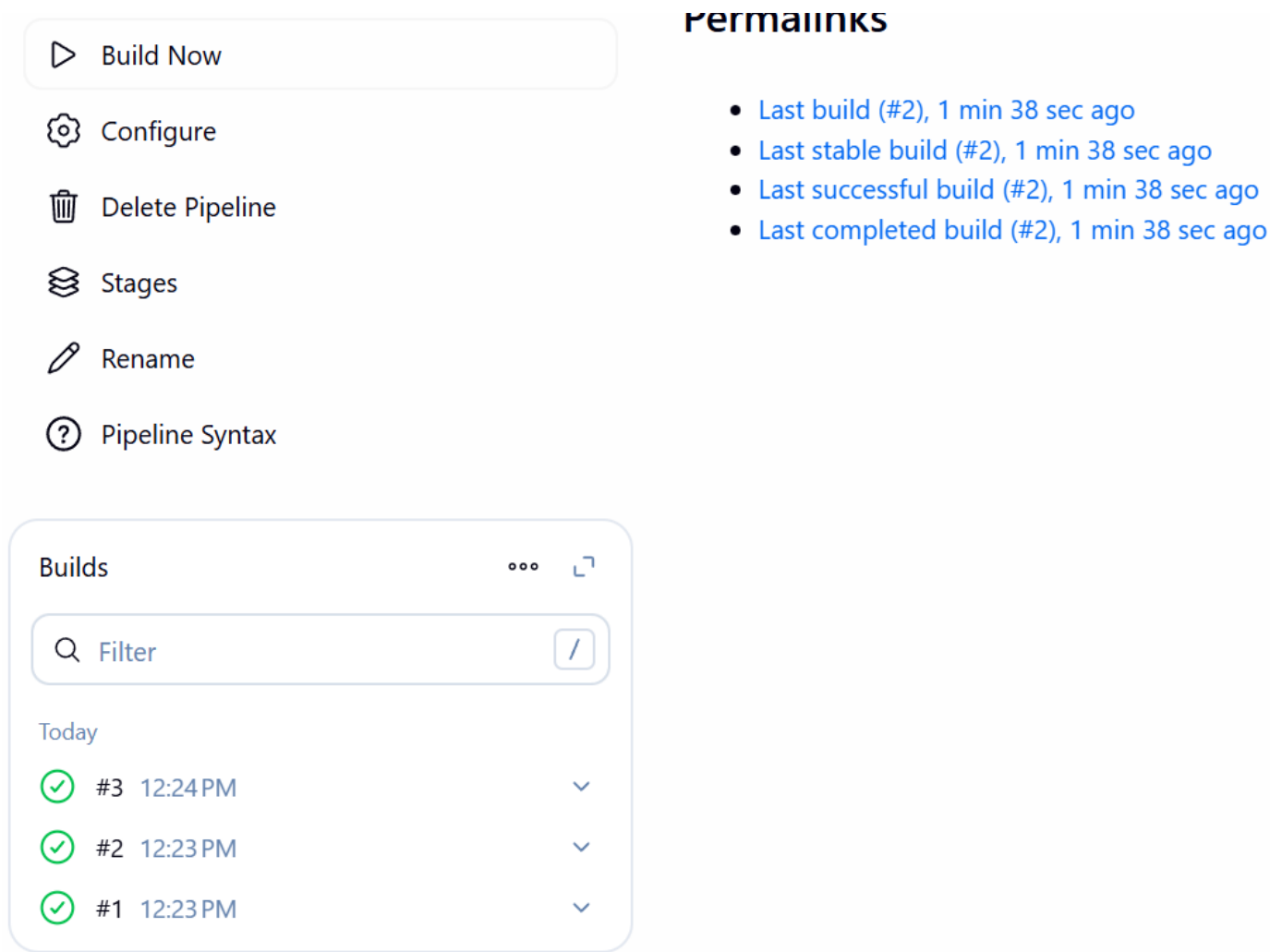
Pipeline script

Script ?

try sample Pipeline...

```
5   }
6   stages {
7   stage('Checkout Git') {
8   steps {
9   git branch: 'main',
10  url: 'https://github.com/swetab-max/SIMPLE-MAVENAPP2.git'
11  }
12  }
13  stage('Build and Test') {
14  steps {
15  bat 'mvn clean test'
16  }
17  }
18  }
19 }
```

CLICK APPLY AND SAVE, IN THE NEXT SCREEN CLICK ON BUILD NOW



CONSOLE OUTPUT:

```
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.067 s -- in com.example.AppTe
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  3.042 s
[INFO] Finished at: 2026-02-12T12:25:07+05:30
[INFO] ------------------------------------------------------------------------
[Pipeline] }
```

File Edit Selection View Go Run Terminal Help

simple-maven-application

EXPLORER

SIMPLE-MAVEN-APPLICATION

- .github\java-upgrade
  - 20260211102252
  - 20260211102451
    - logs
    - plan.md
    - progress.md
  - .gitignore
- src
  - main\java\com\example
    - App.java
  - test\java\com\example
    - AppTest.java
- target
- pom.xml M

Preview plan.md | App.java | plan.md

Upgrade Plan

Project Information

- **Project path**: c:\Users\sweta\Music\simple-maven-application
- **Working branch**: appmod/java-upgrade-20260211102451
  - Note: There are uncommitted changes in the project, which will be stashed before upgrading according to user setting "appModernization.uncommittedChangesAction".
- **Java version**: 11

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell
powershell

```
PS C:\Users\sweta\Music\simple-maven-application> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:simple-maven-app >-------------------
[INFO] Building simple-maven-app 1.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ simple-maven-app ---
[INFO] Deleting C:\Users\sweta\Music\simple-maven-application\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ simple-maven-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\sweta\Music\simple-maven-application\src\main\resources
```